Unit –IV SEMICONDUCTING MATERIALS

INTRODUCTION

We know that materials are classified on the basis of electrical conductivity or resistivity. Semiconductors are materials which have conductivity or resistivity in-between conductors and insulators. The resistivity of a semiconductor is in the range 10^{-4} to 1ohm-metre.

It is not that resistivity alone decides whether a substance is a semiconductor (or) not, because some alloys have resistivity which are in the range of semiconductor's resistivity. Hence, there are some additional properties like band gap, which distinguish the materials as conductors, semiconductors and insulators.

A semiconductor is a solid which has an energy band structure similar to that of an insulator. It acts as an insulator at absolute zero and as a conductor at high temperatures and in the presence of impurities.

PROPERTIES OF SEMICONDUCTORS:

1. The resistivity of semiconductors lies between conducting and insulating materials.
   (i.e.,) 10^{-4} to 1 ohm-metre.

2. In conductors the electrons are charge carriers and will take part in conduction but in semiconductors both electrons and holes are charge carriers and will take part in conduction. (Hole is produced by the vacancy of electron at bond sites)

   Total conductivity $\boldsymbol{\sigma = \sigma_e + \sigma_h}$

   Where, σ_e – conductivity due to electrons.

   σ_h - conductivity due to holes.

3. At 0 K they behave as insulators.
4. When the temperature is raised or when impurities are added, their conductivity increases.

   i.e., $\boldsymbol{\rho \propto 1/T}$

   (This behaviour is opposite compared to usual conductors for which $\boldsymbol{\rho \propto T}$)

5. They have negative temperature coefficient of resistance.

EXPLANATION

The reason for the above property is, when the temperature is increased, large number of charge carriers are produced due to breaking of covalent bonds and hence these electrons move freely and give rise to conductivity.

ELEMENTAL AND COMPOUND SEMICONDUCTORS

Based on the composition of semiconductors they are classified as

(i) Elemental semiconductors.
(ii) Compound semiconductors.

(i) **Elemental semiconductors**: An elemental semiconductor is made of single element from the fourth column elements. Germanium and Silicon are the important example for elemental semiconductors. These are also known as indirect band gap semiconductors. Here, the recombination of an electron from the conduction band with a hole in the valence band takes place via traps. In this process, the phonons are emitted while recombination and they heat the lattice.

(ii) **Compound semiconductors:** A compound semi-conductor is made by combining the third and fifth column elements (or) the second and sixth column elements. GaAs, InP are the important examples for compound semiconductors. These are also known as direct band gap semiconductors. Here, the recombination of electron and hole takes place directly and the energy difference is emitted in the form of photons in the visible (or) infrared range.

Since the life time of the charge carrier is so small, the current amplification is small. Hence these diodes are not suitable for making transistors and ICs; rather they are used in making LEDs and LASER diodes.

DIFFERENCE BETWEEN ELEMENTAL AND COMPOUND SEMICONDUCTOR

S.No.	Elemental Semiconductor (or) Indirect band gap semiconductor	Compound semiconductors (or) Direct band gap semiconductors
1.	They are made of single element eg: Ge, Si.	They are made of compounds. eg: GaA, GaP, CdS, MgO .
2.	They are called **as Indirect band gap semiconductors**. I.e., electron-hole recombination takes place through traps, which are present in the band gap.	They are called as **direct bandgap semiconductors.** i.e., electron-hole recombination takes place directly with each other
3.	Here heat is produced due to recombination.	Here the photons are emitted during recombination (this effect is in LED).
4.	Life time of charge carriers is more due to indirect recombination.	Life time of charge carrier is less due to direct recombination.
5.	Current amplification is more.	Current amplification is less.
6.	They are used for the manufacture of diode and transistors, etc.	They are used for making LED's Laser diode, IC's etc.

Note: At present Germanium and Silicon are the most widely used semiconducting materials.

TYPES OF SEMICONDUCTORS

Based on the purity, semiconductors are classified in to following two types; viz

(i) Intrinsic semiconductors
(ii) Extrinsic semiconductors

INTRINSIC SEMICONDUCTORS

A semiconductor in an extremely pure form is called as intrinsic semiconductors.

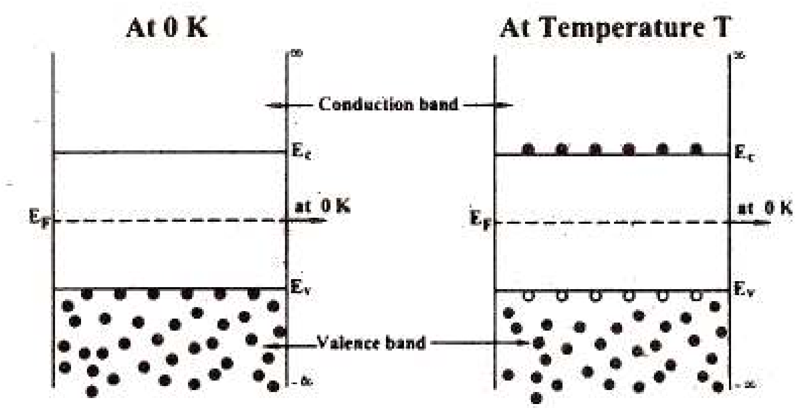
E.g. Germanium and Silicon

In these semiconductors the electrons and holes can be created only by thermal agitation. As there are no impurities the number of free electrons must be equal to the number of holes.

At 0 K the valence band is completely filled and the conduction band is empty. The carrier concentration (i.e.,) electron density (or) hole density increases exponentially with the increase in temperature.

CARRIER CONCENTRATION IN INTRINSIC SEMICONDUCTORS

At 0 K an intrinsic or pure semiconductor behaves as an insulator. But as the temperature increases, some electrons move from valence band to conduction band as shown in fig1.0. Therefore both electrons in conduction band and holes in valence band will contribute to electrical conductivity, and hence the carrier concentrations (or) density of electrons (n_e) and holes (n_h) are important quantities. n_e and n_n for intrinsic semiconductor are calculated as follows:



o - holes

Fig. 1.0 •- electrons

Assume that electron in the conduction band is a free electron of mass m_e and the hole in the valence band is a free particle of mass m_h. The electrons in the conduction band have energies lying from E_c to ∞ and holes in the valence band have energies from $-\infty$ to E_v as shown in fig. Here E_c represent the energy of the bottom (or) lowest level of conduction band and E_v represent the energy of the top (or) the highest level of the valence band.

DENSITY OF ELECTRONS IN CONDUCTION BAND

$$\left.\begin{array}{l}\text{Density of electrons}\\ \text{in conduction band}\end{array}\right\} n_e = \int_{E_c}^{\infty} Z(E) \cdot F(E)\, dE \qquad ...(1)$$

From Fermi-dirac statistics we can write

$$Z(E)\, dE = 2 \cdot \frac{\pi}{4}\left[\frac{8m_e^*}{h^2}\right]^{3/2} E^{1/2}\, dE \qquad ...(2)$$

Considering minimum energy of conduction band as E_c and the maximum energy can go upto ∞ we can write equation (2) as

$$Z(E)\, dE = \frac{\pi}{2} \cdot \left[\frac{8\, m_e^*}{h^2}\right]^{3/2} \cdot (E - E_c)^{1/2}\, dE \qquad ...(3)$$

We know, Fermi function, probability of finding an electron in a given energy state is

$$F(E) = \frac{1}{1 + e^{(E - E_F)/K_B T}} \qquad ...(4)$$

Substituting equation (4) and (3) in equation (1) we have Density of electrons is conduction band within the limits E_c to ∞ as

$$n_e = \frac{\pi}{2}\left[\frac{8m_e^*}{h^2}\right]^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}}{1 + e^{(E - E_F)/K_B T}} \cdot dE \qquad ...(5)$$

Since to move an electron from valence band to conduction band the energy required is greater than $4K_B$ T. (i.e.,) $E - E_F >> K_B T$ (or) $(E - E_F)/K_B T >> 1$

$$\text{(or)} \quad e^{(E - E_F)/K_B T} >> 1$$

$\therefore 1 + e^{(E - E_F)/K_B T} \simeq e^{(E - E_F)/K_B T}$

$\therefore$ Equation (5) becomes

$$n_e = \frac{\pi}{2}\left\{\frac{8m_e^*}{h^2}\right\}^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}}{e^{(E - E_F)/K_B T}}\, dE$$

$$\text{(or)} \qquad n_e = \frac{\pi}{2} \cdot \left\{\frac{8m_e^*}{h^2}\right\}^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} \cdot e^{(E_F - E)/K_B T}\, dE \qquad ...(6)$$

Let us assume that $E - E_c = x K_B T$

$$\text{(or)} \qquad E = E_c + x K_B T$$

Differentiating we get $dE = K_B T \cdot dx$,

Limits: When $E = E_c$; $x = 0$

When $E = \infty$; $x = \infty$

$\therefore$ **Limits are 0 to ∞**

$\therefore$ Equation (6) can be written as

$$n_e = \frac{\pi}{2} \cdot \left\{\frac{8m_e^*}{h^2}\right\}^{3/2} \int_0^\infty (xK_BT)^{1/2} \cdot e^{(E_F - xK_BT - E_c)/K_BT} K_BT\, dx$$

$$n_e = \frac{\pi}{2} \cdot \left\{\frac{8m_e^*}{h^2}\right\}^{3/2} \int_0^\infty (x)^{1/2} (K_BT)^{3/2} \cdot e^{(E_F - E_c)/K_BT} e^{-x}\, dx$$

$$n_e = \frac{\pi}{2} \cdot \left\{\frac{8m_e^* K_BT}{h^2}\right\}^{3/2} e^{(E_F - E_c)/K_BT} \int_0^\infty x^{1/2} \cdot e^{-x}\, dx$$

$$n_e = \frac{\pi}{2} \cdot \left\{\frac{8m_e^* K_BT}{h^2}\right\}^{3/2} e^{(E_F - E_c)/K_BT} \frac{\sqrt{\pi}}{2} \qquad \left[\text{since} \int_0^\infty x^{1/2} e^{-x}\, dx = \frac{\sqrt{\pi}}{2}\right]$$

$$n_e = \frac{1}{4}\left[\frac{8\pi m_e^* K_BT}{h^2}\right]^{3/2} e^{(E_F - E_c)/K_BT}$$

$\therefore$ Density of electrons in conduction band is

$$\boxed{n_e = 2\left\{\frac{2\pi m_e^* K_B T}{h^2}\right\}^{3/2} \cdot e^{(E_F - E_c)/K_BT}} \qquad \text{...(7)}$$

DENSITY OF HOLES IN VALENCE BAND

We know, F(E) represent the probability of filled state. As the maximum probability will be 1, the probability of unfilled states will be [1-F(E)].

Example, if F(E) = 0.8 then 1-F(E) = 0.2

i.e., 80% change of finding an electron in valence band and 20% change of finding a hole in valence band.

Let the maximum energy in valence band be E_v and the minimum energy be $-\infty$, Therefore density of hole in valence band n_h is given by

$$n_h = \int_{-\infty}^{E_v} Z(E)\,[1 - F(E)]\, dE \qquad \text{...(8)}$$

We know $Z(E)\,dE = \frac{\pi}{2}\left\{\frac{8m_h^*}{h^2}\right\}^{3/2} (E_v - E)^{1/2}\,dE$ ...(9)

$$1 - F(E) = 1 - \frac{1}{1 + e^{(E - E_F)/K_BT}}$$

$$= \frac{e^{(E - E_F)/K_BT}}{1 + e^{(E - E_F)/K_BT}}$$

Here $E - E_F << K_BT$,

$$\text{(or)} \quad \frac{E - E_F}{K_BT} << 1 \qquad \therefore e^{(E - E_F)/K_BT} << 1$$

$$\text{(or)} \quad 1 + e^{(E - E_F)/K_BT} \simeq 1$$

$$\therefore 1 - F(E) = e^{(E - E_F)/K_BT} \qquad \text{...(10)}$$

substituting equation (10) and (9) in (8), we get

$$n_h = \frac{\pi}{2} \cdot \left\{\frac{8m_h^*}{h^2}\right\}^{3/2} \int_{-\infty}^{E_v} (E_v - E)^{1/2} \cdot e^{(E - E_F)/K_BT}\,dE \qquad \text{...(11)}$$

Let us assume that $E_v - E = xK_BT$, (or) $E = E_v - xK_BT$

differentiating we get $dE = -K_BT\,dx$

Limits: When $E = -\infty$

We have $E_v - (-\infty) = x$

$$\therefore x = \infty$$

When $E = E_v$; $x = 0$

$\therefore$ Limitsare ∞ to 0

$\therefore$ Equation (11) becomes

$$n_h = \frac{\pi}{2} \cdot \left\{\frac{8m_h^*}{h^2}\right\}^{3/2} \int_{\infty}^{0} (xK_BT)^{1/2} \cdot e^{(E_v - xK_BT - E_F)/K_BT} (-K_BT)dx$$

To exclude the negative sign, the limits can be interchanged.

$$\therefore n_h = \frac{\pi}{2}\left[\frac{8m_h^*}{h^2}\right]^{3/2} \int_0^{\infty} x^{1/2} (K_BT)^{3/2} e^{(E_v - E_F)/K_BT} \cdot e^{-x} \cdot dx$$

$$n_h = \frac{\pi}{2} \cdot \left\{\frac{8m_h^* K_BT}{h^2}\right\}^{3/2} e^{(E_v - E_F)/K_BT} \int_0^{\infty} x^{1/2} \cdot e^{-x}\,dx$$

$$n_h = \frac{\pi}{2} \cdot \left\{ \frac{8m_h^* K_B T}{h^2} \right\}^{3/2} e^{(E_v - E_F)/K_B T} \frac{\sqrt{\pi}}{2} \qquad \left[\text{since} \int_0^\infty x^{1/2} e^{-x}\, dx = \frac{\sqrt{\pi}}{2} \right]$$

$$n_h = \frac{1}{4} \left[\frac{8\pi m_h^* K_B T}{h^2} \right]^{3/2} e^{(E_v - E_F)/K_B T}$$

$\therefore$ Density of holes in valence band is

$$\boxed{n_h = 2 \left\{ \frac{2\pi m_h^* K_B T}{h^2} \right\}^{3/2} e^{(E_v - E_F)/K_B T}} \qquad \text{...(12)}$$

VARIATION OF FERMI ENERGY LEVEL AND CARRIER CONCENTRATION WITH TEMPERATURE IN AN INTRINSIC SEMICONDUCTOR

For an intrinsic semiconductor, number of electron per unit Volume or electron density will be the same as the hole density.

(i.e.,) $n_e = n_h$

Equating eqn (7) and (12), we can write

Equating equations (7) and (12), we can write

$$\left(m_e^* \right)^{3/2} e^{(E_F - E_c)/K_B T} = \left(m_h^* \right)^{3/2} e^{(E_v - E_F)/K_B T}$$

$$\left\{ \frac{m_h^*}{m_e^*} \right\}^{3/2} = \frac{e^{(E_F - E_c)/K_B T}}{e^{(E_v - E_F)/K_B T}}$$

$$= e^{(E_F - E_c - E_v + E_F)/K_B T}$$

$$\text{(or)} \quad \left(\frac{m_h^*}{m_e^*} \right)^{3/2} = e^{(2E_F - (E_v + E_c))/K_B T}$$

Taking log on both sides we have

$$3/2 \log \left(\frac{m_h^*}{m_e^*} \right) = \frac{[\, 2E_F - (E_v + E_c) \,]}{K_B T}$$

$$2E_F = E_c + E_v + \frac{3}{2} K_B T \log \frac{m_h^*}{m_e^*}$$

$$\boxed{E_F = \frac{E_c + E_v}{2} + \frac{3}{4} K_B T \log\left(\frac{m_h^*}{m_e^*}\right)} \qquad ...(13)$$

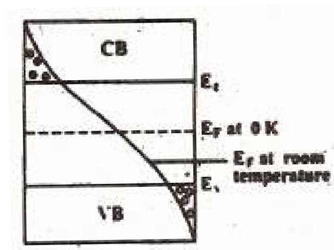
If $m_h^* = m_e^*$, then $\log \frac{m_h^*}{m_e^*} = 0 \quad [\because \log 1 = 0]$

$\therefore$ Equation (13) becomes

$$\boxed{E_F = \frac{(E_c + E_v)}{2}} \qquad ...(14)$$

i.e., the Fermi energy level lies in the midway between E_c and E_v as shown in fig 1.1(since at 0 K, T = 0).
But in actual case $\mathbf{m_h > m_e}$ and the Fermi energy level slightly increases with the increase in temperature as shown in fig1.1.

Fig 1.1



o - holes

•- electrons

EXTRINSIC SEMICONDUCTORS

Impure semiconductors in which the charge carriers are produced due to impurity atoms are called extrinsic semiconductors. They are obtained by doping intrinsic semiconductors with impurity atoms.

Based on the type of impurity added they are classified into

(I) n- type semiconductors
(II) p- type semiconductors

(I) n-type semiconductors: n-type semiconductors is obtained by doping an intrinsic semiconductor with pentavalent (5 electrons in valence band) impurity atoms like phosphorus, arsenic, antimony, etc.,

The 4 valence electrons of the impurity atoms bond with 4 valence electrons of the semiconductor atom and remaining 1 electron of the impurity atom is left free as shown in fig1.3 .

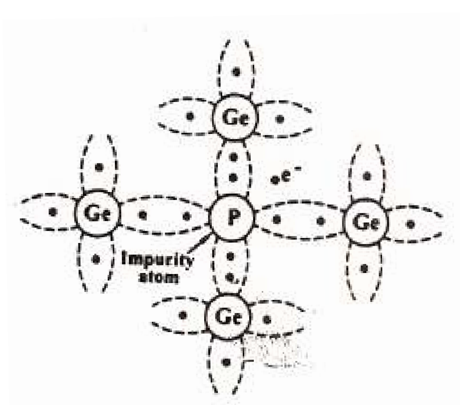


Fig 1.3

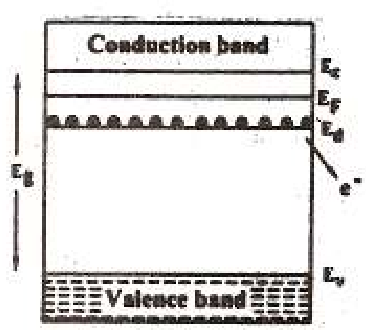


Fig 1.4

Therefore number of free electrons increases. As the electrons are produced in excess, they are the majority charge carriers in n-type semiconductors and holes are the minority charge carriers. Since electrons are donated in the type of semiconductors the energy level of these donated electrons is called donor energy level (E_d) as shown in fig 1.4 E_d is very close to conduction band and hence even at room temperature the electrons are easily excited to conduction band. The current flow in this type of semiconductors is due to electrons.

(ii) p-type semiconductors:

p-type semiconductor is obtained by doping an intrinsic semiconductor with trivalent (3 electrons in valence band) impurity atoms like Boron, Gallium, Indium, etc.

The three valence electrons of the impurity atom pairs with three valence electrons of semiconductor atom and one position of the impurity atom remains vacant, this is called hole as shown in fig 1.5

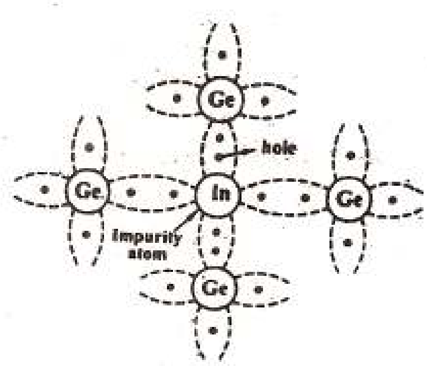


Fig 1.5

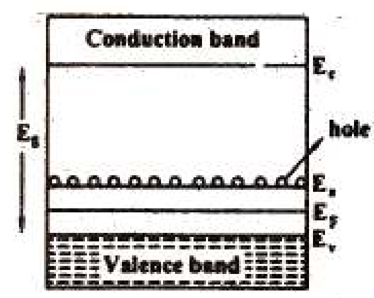


Fig 1.6

Therefore the number of holes is increased with the impurity atoms added to it. Since holes are produced in excess, they are the majority charge carriers in p-type semiconductors and electrons are the minority charge carriers.

Since the impurity can accept the electrons this energy level is called acceptor energy level (E_a) and is present just above the valence band as shown in fig 1.6. Here, the current conduction is mainly due to holes (holes are shifted from on covalent bond to another)

HALL EFFECT

Measurement of conductivity alone can not determine whether the conduction is due to electron or holes and therefore will not distinguish between p-type and n-type semiconductor.

Therefore Hall Effect is used to distinguish between the two types of carriers and their carrier densities and is used to determine the mobility of charge carriers.

Definition:

When a conductor (metal of semiconductor) carrying a current is placed in a transverse magnetic field and electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. The phenomenon is known as Hall effect and the generated voltage is called "**Hall Voltage**".

Hall Effect in n-type semiconductor

Let as consider an n-type material to which the current is allowed to pass along x-direction from left to right and the magnetic field is applied in z-direction, as a result Hall voltage is produced in y direction as shown in fig 1.7

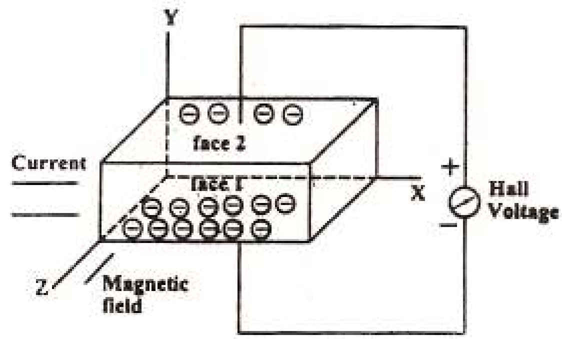


Fig 1.7

Since the direction of current is from left to right the electrons move form right to left in x-direction as shown in fig 1.7 and Fig 1.8

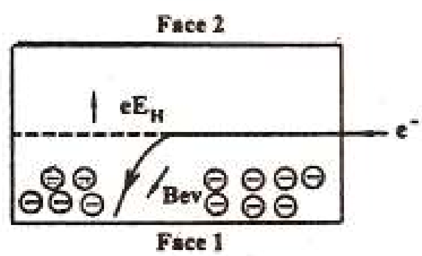


Fig 1.8

Now due to the magnetic field applied the electrons move towards downward direction with the velocity 'v' and cause the negative charge to accumulate at face (1) of the material as shown in fig. Therefore a potential difference is established between face (2) and face (1) of the specimen which gives rise to field E_H in negative y direction.

Here, the Force due to potential difference $= -eE_H$ ...(1)

Force due to magnetic field $= -Bev$ ...(2)

$\therefore$ At equilibrium equation (1) = equation (2)

$$-eE_H = -Bev$$

$$\text{(or) } E_H = Bv \qquad ...(3)$$

We know the current density J_x in the x direction is

$$(\text{or})\ v = -\frac{J_x}{n_e e} \quad ...(4)$$

Substituting equation (4) in equation (3) we get

$$E_H = -\frac{BJ_x}{n_e e} \quad ...(5)$$

$$(\text{or}) \quad E_H = R_H \cdot J_x \cdot B \quad ...(6)$$

Where R_H is known as the Hall coefficient, given by

$$\boxed{R_H = -(1/n_e e)}$$

The negative sign indicates that the field is developed in the negative direction.

HALL EFFECT IN P-TYPE SEMICONDUCTOR

Let us consider a p-type material for which the current is passed along x-direction from left to right and magnetic field is applied along z-direction as shown in fig 1.9. Since the direction of current is from left to right, the holes will also move in the same direction as shown in fig 1.10.

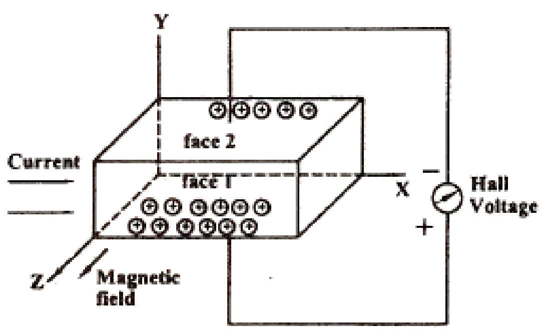


Fig 1.9

Now due to the magnetic field applied, the holes move toward the downward direction with velocity 'v' and accumulate at the face (1) as shown in fig 1.9 &1.10

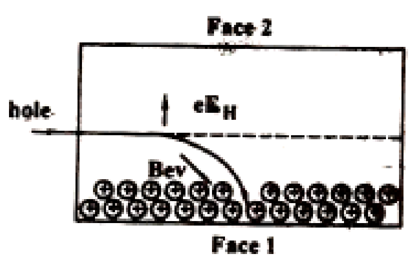


Fig 1.10

A potential difference is established between face (1) and (2) in the positive y direction.

Force due to the potential difference = eE_H ….. (7)

Force due to magnetic field = Bev ……. (8)

(since hole is considered to be an electron with same mass but positive charge negative sign in not included)

At equilibrium equation (7) = equation (8)

Therefore $eE_H = Bev$

(or) $E_H = Bv$ ……… (9)

We know current density $J_x = n_h ev$

(or) $v = (J_x/n_h)e$ ……… (10)

Where n_h is the hole density.

Substituting equation (10) in (9) we get

$$E_H = \frac{BJ_x}{n_h e}$$

(or) $E_H = R_H J_x B$

Where $\boxed{R_H = \frac{1}{n_h e}}$ ………(11)

Equation (11) represents the Hall coefficient and the positive sign indicates that the Hall field is developed in the positive y direction.

HALL COEFFICIENT INTERMS OF HALL VOLTAGE

If the thickness of the sample is t and the voltage developed is V_H, then

Hall voltage $V_H = E_H \cdot t$ ...(12)

Substituting equation (6) in equation (12), we have

$$V_H = R_H J_x B \cdot t \quad ...(13)$$

If b is the width of the sample then

Area of the sample = b.t.

$\therefore$ Current density $= J_x = \frac{I_x}{bt}$ ...(14)

Substituting equation (14) in equation (13) we get

$$V_H = \frac{R_H I_x B t}{bt}$$

$$\text{(or)} \quad V_H = \frac{R_H I_x B}{b}$$

$$\boxed{\therefore \textbf{Hall coefficient, } R_H = \frac{V_H b}{I_x B}}$$

Note: The sign for V_H will be opposite for n and p type semiconductors

EXPERIMENTAL DETERMINATION OF HALL EFFECT

A semiconductor slab of thickness'‘t’ and breadth ‘b’ is taken and current is passed using the battery as shown in fig1.11

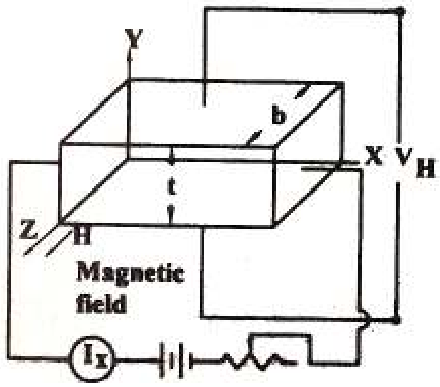


Fig 1.11

The slab is placed between the pole pieces of an electromagnet so that current direction coincides with x-axis and magnetic field coincides with z-axis. The Hall voltage (v_H) is measured by placing two poles at the center of the top and bottom faces of the slab (y-axis).

It is a magnetic field applied and the V_B is the Hall voltage produced, then the Hall coefficient can be calculated from the formula

$$R_H = \frac{V_H\, b}{I_x\, B}$$

MOBILITY OF CHARGE CARRIERS

In general the Hall Co-efficient can be written as $\mathbf{R_H} = \frac{1}{ne}$ $n_h e$

The above expression is valid only for conductors where the velocity is taken in drift velocity.

But for semiconductors velocity is taken as average velocity so R_H for an ‘n’ type semiconductor is modified as

$$R_H = -\frac{3\pi}{8}\left(\frac{1}{n_e\, e}\right)$$

$$R_H = \frac{-1.18}{n_e\, e} \qquad ...(1)$$

We know the conductivity for n type is $\sigma_e = n_e\, e\, \mu_e$

$$\text{(or)}\ \mu_e = \frac{\sigma_e}{n_e\, e} \qquad ...(2)$$

From equation (1) we can write

$$\frac{1}{n_e\, e} = \frac{-R_H}{1.18} \qquad ...(3)$$

Substituting equation (3) in equation (2) we get

$$\mu_e = -\frac{\sigma_e R_H}{1.18}$$

∴ The *mobility of electron is in an n-type semiconductor* is

$$\boxed{\mu_e = \frac{-\sigma_e V_H b}{1.18 I_x B}} \quad \left[\because R_H = \frac{V_H b}{I_x B}\right]$$

Similarly for p-type semiconductor the mobility of hole is

$$\boxed{\mu_h = \sigma_h \cdot \frac{V_H b}{1.18 I_x B}}$$

Thus by finding Hall voltage, Hall coefficient can be calculated and thus then the mobility of the charge carriers (μ_e and μ_h) can also be determined.

Hall Angle (θ_H)

We know if the current (E_x) is applied to a specimen along x direction, magnetic field (B) along z-direction the Hall voltage (E_H) is produced along y-direction as shown in fig 1.12

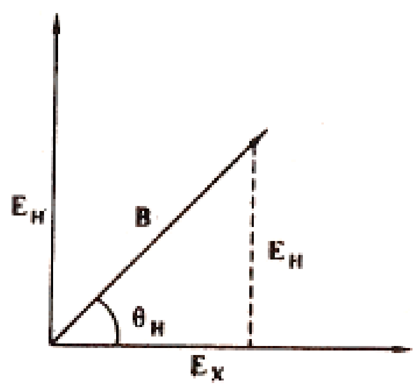


Fig 1.12

Here the Hall angle can be measured from the formula

If the thickness of the sample is t and the voltage developed is V_H, then

Hall voltage $V_H = E_H \cdot t$ ...(12)

Substituting equation (6) in equation (12), we have

$$V_H = R_H J_x B \cdot t \qquad ...(13)$$

If b is the width of the sample then

Area of the sample $= b.t.$

$\therefore$ Current density $= J_x = \frac{I_x}{bt}$ ...(14)

Substituting equation (14) in equation (13) we get

$$V_H = \frac{R_H I_x B t}{bt}$$

$$\text{(or)} \quad V_H = \frac{R_H I_x B}{b}$$

$$\boxed{\therefore \text{ Hall coefficient, } R_H = \frac{V_H b}{I_x B}}$$

$$\tan \theta_H = \frac{E_H}{E_x} \quad ...(1)$$

we know, $E_H = R_H J_x B$ ...(2)

Also we know $R_H = -1/n_e e$, and $J_x = -n_e e v_x$

where $v_x \rightarrow$ velocity of the electron

$\therefore$ Substituting these values in equation (2) we get

$$E_H = \left(\frac{-1}{n_e e}\right) \cdot (-n_e e v_x) \cdot B$$

$$\text{(or)} \quad E_H = v_x \cdot B$$

Substituting equation (3) in equation (1) we get

$$\tan \theta_H = \frac{v_x \cdot B}{E_x}$$

$$\textbf{(or)} \quad \boxed{\tan \theta_H = \mu B}$$

Where $\mathbf{\mu = V_x/E_x}$. Thus the mobility **(μ)** can be defined as the velocity acquired by the charge carrier per unit electric field.

APPLICATIONS OF THE HALL EFFECT

(i) It is used to determine whether the material is p-type or n-type semiconductor. (i.e.,) if $R_H$is positive then the material is p-type.

(ii) It is used to find the carrier concentration, $n_e= - (1/eR_H)$ and $n_p=(1/eR_H)$

(iii) It is used to find the mobility of charge carriers μ_e , μ_h

(iv) It is used to determine the sign of the current carrying charges.

(v) It is used to design magnetic flux meters and multipliers on the basis of Hall Voltage.

(vi) It is used to find the power flow in an electromagnetic wave.

SUPERCONDUCTING MATERIALS

INTRODUCTION

Before the discovery of super conductivity, it was thought that the electrical resistance of a conductor becomes zero only at absolute zero. But, it was found that in some materials, the electrical resistance becomes zero, when they are cooled to very low temperatures.

For example, the electrical resistance of pure mercury suddenly drops to zero, when it is cooled below 4.2 Kelvin and becomes a superconductor. This phenomenon was first observed by H. Kamerlingh Onnes in 1911. This phenomenon of losing the resistivity by a solid absolutely, when cooled to sufficiently low temperature, is called **superconductivity.**

TRANSITION TEMPERATURE OR CRITICAL TEMPERATURE.

The temperature at which a normal conductor loses its resistivity and becomes a super conductor is known as **transition temperature** or **critical temperature** (T_c) as shown in figure 1.13.

(i) If the transition temperature is low, then the superconductors are known **as low Temperature SuperConductors.**

(ii) If the transition temperature is high >30K then it is **known high temperature superconductor.**

The transition temperature depends on the properties of that material. It is found that the superconducting transition is reversible (i.e.) above critical temperature (T_c), the super conductor again becomes normal conductor. Also T_c is defined for any particular materials.

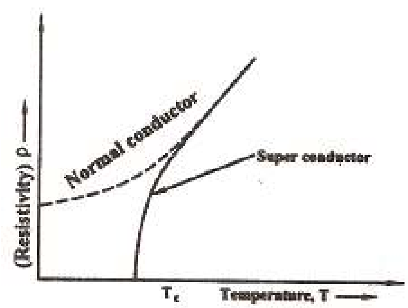


Fig 1.13

TEMPERATURE DEPENDENCE OF RESISTANCE AND SUPER CONDUCTING PHENOMENA

We know conduction in material is due to flow of electrons, which are loosely bound to the outer most orbit in an atom. If a small amount of energy (thermal energy) is supplied to this atom, the free electrons become conduction electrons. These electrons leave the atoms to which they were originally bound and the atoms become positive ion cores.

Due to thermal excitation these positive ions vibrate in equilibrium positions. These vibrations are called **Lattice Vibrations**. The quanta of energy which are emitted during these vibrations are called Phonons.

In metals the electrical resistivity (ρ) is due to

1. Presence of impurities in it (ρ_0)
2. Lattice vibrations, which depends on temperature [ρ (T)]

i.e. According to **Mathiessen's Rule**

$$\rho = \rho_0 + \rho\,(T)$$

Where, $\rho_0 \rightarrow$ Residual resistivity due to scattering by impurities.

ρ(T)$\rightarrow$ *Ideal* resistivity due to scattering by phonons.

In both the above said cases, the moving electrons will be SCATTERED and this gives rise to electrical resistivity. The electrical resistivity at high temperature and low temperature are discussed below.

At high temperature

Case (i) Impure metals: In the case of impure metals the resistivity exists due to impurities and also due to phonon vibrations (due to increase in temperature) as shown in fig1.14.

$$\rho - \rho_0 + \rho\,(T)$$

Case (ii) Pure metals: When a pure metal is taken, the resistivity due to impurities are minimized. Still, due to higher temperature, scattering by phonons act as resistance and hence. $\rho = \rho$ **(T).** Which increases linearly with temperature as shown in the above fig 1.14.

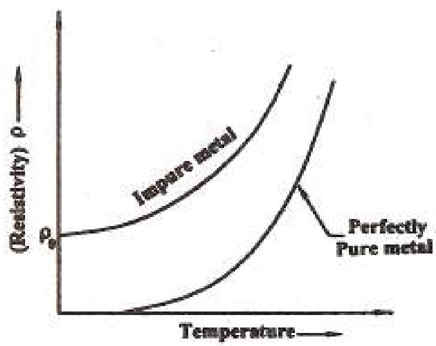


Fig 1.14

At low temperature

Case (i) Impure metals: When the temperature is reduced to 0 K, the resistivity of the impure metal doesn't becomes zero, because there exists some impurities which give rise to minimum resistivity known as residual resistivity ρ_0 as shown in the above fig 1.14

$$\rho = \rho_0 \text{ [since } \rho\text{ (T) =0]}$$

Case (ii) Pure metals: In a pure metal, where the temperature is lowered the electrons are ordered in a particular direction. The resistivity due to both the impurities and phonon vibrations are absent and hence the resistivity $\rho = 0$. **Hence the material becomes a super conductor,** below the critical temperature as shown in fig 1.14.

Thus at low temperature the electrons moves freely over the lattice points and have very less vibrations giving rise to the phenomenon of **superconductivity.**

PROPERTIES OF SUPER CONDUCTORS

1. **Electrical Resistance:**

The electrical resistance of a superconducting material is experimentally found to be very low << 10^{-8} **ohm.cm** and hence theoretically ρ is taken as zero.

2. Magnetic property:

When super conducting materials are subjected to very large value of magnetic fields, the super conducting property is destroyed.

The filed required to destroy the superconducting property is called as critical magnetic field (H_c) given as

$$H_c = H_0\left[1 - \frac{T^2}{T_c^2}\right]$$

Where , H_0 - Critical field at 0 K [A/m]

T_c - Transition temperature.

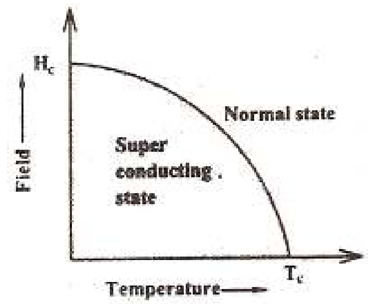


Fig 1.15

From the fig 1.15 we can find that when the temperature of the material increases, the value of the critical magnetic field decreases. Also the value of the critical field will be different for different materials

3. Diamagnetic property – MEISSNER EFFECT

When a superconducting material is placed in a magnetic field of flux density 'B' the magnetic lines of force penetrate through the material as shown in fig 1.16.

Now, when the material is cooled below its transition temperature (i.e.), when $T \leq T_c$ then the magnetic lines of force are expelled (or) ejected out from the material as shown in fig 1.17

We know that a diamagnetic material have the tendency to expel the magnetic lines of force. Since the superconductor also expels the magnetic lines of forces is behaves as perfect diamagnet. This behavior was first observed by Meissner and is hence called as the Meissner effect.

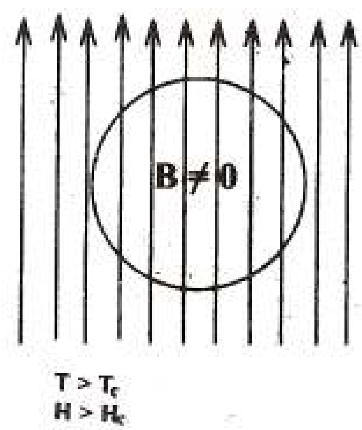


Fig 1.16

Meissner effect: When a superconducting material is placed in a magnetic field, under the condition when $T \leq T_c$ and $H \leq H_c$ the flux lines are excluded from the material. Thus the material exhibits perfect diamagnetism. This phenomenon is called as Meisser effect.

Proof:

We Know , $B = \mu_0 (H + M)$

When $B = 0$ We get $0 = \mu_0 (H + M)$

Since $\mu_0 \neq 0$ We can write $H + M = 0$

(or) $-H = +M$

(or) $\frac{M}{H} = -1 = \chi$

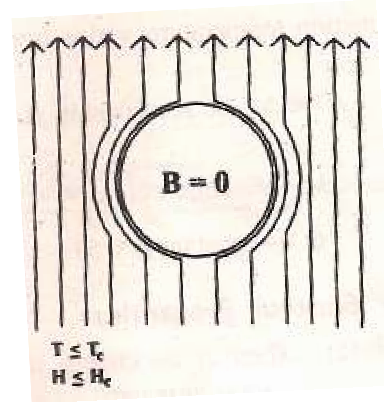


Fig 1.17

The susceptibility is negative, indicating that superconductor is a perfect diamagnet.

4. Effect of electrical current

When a large value of current is applied to a super conducting material it induces some magnetic field in the material and because of this magnetic field, the super conducting property of the material is destroyed.

5. Persistant Current

When d.c. Current of large magnitude is once induced in a super conducting ring then the current persists in the ring even after the removal of the field as shown in fig 1.18. This is known as **persistant Current**. This is due to the diamagnetic property (i.e.) the magnetic flux inside the ring will be trapped in it and hence the current persists.

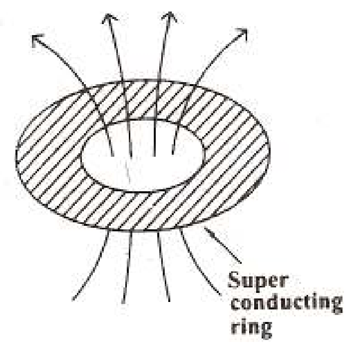


Fig 1.18

6. Thermal properties

(i) The internal energy and specific heat decrease at transition temperature

(ii) The thermal conductivity of type I superconductor is low.

(iii) The thermo-electric effect disappears in the superconducting state.

7. ISOTOPE EFFECT

For a given superconductor the transition temperature varies with the mass of the isotope. The dependence of T_c on the atomic mass M is given by

$$T_C \ \alpha \ \frac{1}{M^{\alpha}}$$

Where M- atomic weight and the exponent α is close to 0.5 for most superconductors

8. General Properties

(a) There is no change in elastic properties, photo electric properties and crystal structure.

(b) The transition temperature is changed with the frequency variation.

TYPES OF SUPER CONDUCTOR

There are two types of superconductors based on their variation in magnetization, due to external magnetic field applied. Viz,

(i) Type I super conductor (or) Soft super conductor
(ii) Type II super conductor (or) Hard super conductor

Type I (soft) Super conductor

When a superconductor is kept in the magnetic field and if the field is increases the super conductor becomes a normal conductor abruptly at a critical magnetic field as shown in figure 1.19. This type of materials is termed as **Type I superconductors.**

Below critical field, the specimen excludes all the magnetic lines of force and exhibits perfect Meissner effect. Hence Type I superconductors are perfect diamagnets, represented by the negative sign in magnetization (since for diamagnet χ is -ve)

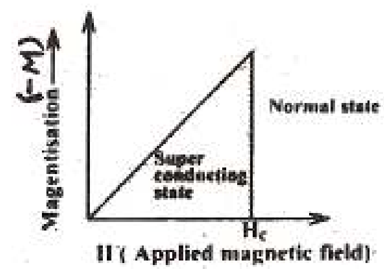


Fig 1.19

Type II (hard) super conductors

When a superconductor is kept in the magnetic field and if the field is increased, below the lower critical field Hc_1, the material exhibits perfect diamagnetism (i.e.) it behaves as a superconductor and above Hc_1, the magnetization decreases and hence the magnetic flux starts penetrating through the material. The specimen is said to be in a mixed state between Hc_1 and Hc_2, Above Hc_2 (upper critical field) it becomes a normal conductor as shown in fig 1.20

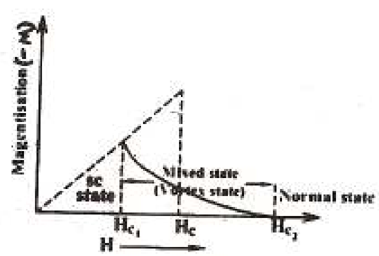
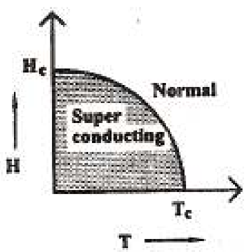
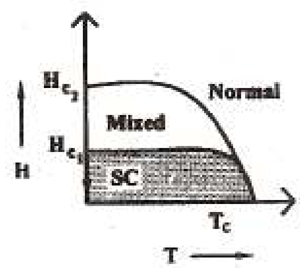


Fig 1.20

The materials which loses their superconducting property gradually due to the increase in magnetic field are called **Type II super conductor.**

DIFFERENCE BETWEEN TYPE I AND TYPE II SUPERCONDUCTORS

S.NO.	TYPE I (SOFT) SUPERCONDUCTOR	TYPE II (HARD) SUPERCONDUCTOR
1.	Type I superconductor becomes a normal conductor abruptly at critical magnetic field	Type II super conductor loses its super conducting property gradually, due to increase in magnetic field.
2.	Here we have only one critical field (H_c).	Here we have two critical field (i.e.) Lower critical fields (H_{c1}) and upper critical field (H_{c2})
3.	No mixed state exists	Mixed (or) Vortex state is present.
4.	Highest known critical field is 0.1 Tesla	The critical field is greater than Type I (i.e.) upto 30 Tesla
5.	Magnetic phase diagram 	Magnetic phase diagram 
6.	Example: Pb, Zn, Al, Ga, Hg	Example:Nb_3Sn, Vanadium, HTSCs

HIGH TEMPERATURE SUPERCONDUCTORS

Intermetallic compounds with A-15 structure have higher superconducting transition temperature upto 25K. Sodium tungsten bronzes and barium lead bismuth oxides are important intermetallic compounds with high superconducting transition temperature (20K). In IBM lab, Switzerland Laboratory, during the end of 1986, Bednorz and Mueller was discovered that a crystalline compound of lanthanum, strontium and copper grown in an atmosphere of oxygen has the highest superconducting transition temperature, 37K. Following this discovery, the Chinese group of scientist has found a superconducting state in an octahedral crystalline of lead, lanthanum, copper and oxygen at 77K.

After that C.W.Chu and his collaborators of University of Houston, U.S.A. found stable superconductivity at 90K in yttrium barium-copper oxide compounds. Detection of superconductivity especially in yttrium complexes at operating temperature much above 130K has been performed last year. Now the stable and reproducing superconducting state in observe at a high temperature around 165K. Researches are going on to get the superconducting state at room temperature conditions.

Some of the high temperature (T_c >100K) superconducting are given below:

$Tl_2 Ba_2 Ca_2 Cu_3 O_{y3}$ $La_{2-x} Sr_x Cuo_4$, $YBa_2 Cu_3 O_{7-y}$ and $Bi_2 (CaSr)_3 Cu_2 O_y$. Here the value of x is less than 0.2. Further the value of y is a variable one and depends upon the stability of the system. The rare earth oxides systems, like La-Sr-Cu-O and Y-Ba-Cu-O have so far been found to have stable superconducting transition temperature in the range 80-100K range.

Recently T.Ogushi and his associates from Japan have reported in La-Sr-Nb-O system (without copper) that shows superconductivity upto 225K. Some scientists believe that this copper less Y-Sr-Nb-O or La-Sr-Nb-O system may lead to the real room temperature superconductors.

New oxide superconductors having T_c > 90K without rare earth element

New oxides superconductors with above 90 K transition temperature are discovered in 1988. These materials do not even give the crystal structures as 1-2-3. The newer compounds are still layered copper oxide materials, but they do not contain rare earth elements. In place of yttrium or other rare earths, the compounds contain metals like bismuth or thallium. Thus they contain four metallic elements instead of three. For example, thallium- barium- calcium- copper oxide superconductor contains three copper oxygen layers and it becomes superconductor at 120K.

Preparation and Characterization of High temperature Ceramic superconductor

The new ceramic oxide superconductors like yttrium barium copper oxide, lanthenam barium copper oxide, Neodymium barium copper oxide and thallium barium copper oxide are all type II superconductors which have higher critical fields and hence are used to manufacture superconducting magnets having higher strengths.

Lets us see the preparation of yttrium barium copper oxide superconductor by the ‘shake and bake’ method. This method involves a four step process:

1. Mixing the chemicals
2. Calcinations (initial firing)
3. The intermediate firing (oxygen annealing) and
4. The final oxygen annealing.

1. Mixing the chemicals

The powders of yttrium oxide, barium carbonate and cupric oxide are taken such that 11.29 gram of Y_2O_3, 39.47 gram of BaCO3 and 23.86 gram of CuO so that the atomic ratios of yttrium, barium and copper are 1:2:3. A mortar with pestle is used to grind down any lumps or large particle in the powders. Then the powder mixture is shaked vigorously in a capped jar or stoppered flask for several minutes.

2. Calcination

After the powders are thoroughly mixed, they are kept in an alumina crucible and heated in a furnace of about 950^0C for about 18 hours. The following chemical reaction takes place.

$$0.5Y_2O_3 + 2\,Ba\,CO_3 + 3\,Cu\,O \longrightarrow Y\,Ba_2\,Cu_3\,O_{6.5} + 2CO_2 \uparrow$$

This initial heat treatment is called calcination and results in a porous dark gray or black the clump and the basic crystal structure is developed. The material formed will shrinked and becomes much denser than the original powder mix.

3. Intermediate firing

The porous black clump is ground into a fine powder and place in the furnace. After the furnace temperature reaches about 500^0C, a slow flow of oxygen into the furnace is maintained. This heat treatment under oxygen flow is called annealing. The oxygen flow should be maintained in a constant rate for 18 hours at 950^0C. The maximum temperature of 950^0C should be reached in an hour. After this firing, the cooling rate must be more than 100^0C per hour upto 400^0C and the oxygen flow is also maintained. By this manner the oxygen content of each crystal unit is increased from 6.5 atoms to 7 atoms and the chemical formula for the material becomes $YBa_2\,Cu_3\,O_7$. At 400^0C, the supply of oxygen is cutoff and should be thoroughly re-ground in a mortar and pestle.

4. The final oxygen annealing

The reground black is placed back in the alumina crucible. The powder should be as finely ground and as densely evenly packed. Now the sample is heated to 950^0C and 1000^0C. For about 18 hours. The cooling take places in a very slow manner with adequate oxygen flow. The rate of cooling must be less than 100^0C per hour upto 400^0C. After that the rate of cooling can be increased.

Perovskite superconductivity

It is found that ceramic superconductors gave the unit cell such that one atom of a rare earth metal, two barium atoms, three copper atoms and seven oxygen atoms. In the superconductor the metallic atoms are in the 1 : 2 : 3. Thus these new kinds of superconductors are called 1-2-3 superconductors. The electrical conductivity of 1-2-3 superconductors is based on the presence of oxygen atoms in their unit cell. The oxygen atoms have more vacancies and so they can accommodate electrons from other atoms. It is found that electricity flow through the material by allowing these holes to move from atom in the crystal.

The unit of $YBa_2 Cu_3 O_9$ is a stack of alternate cubic pervoskite structures as shown in fig1.21 $(BaCuO_3)$: $(Y Cu O_3)$: $(BaCuO_3)$. But is not a superconductor. The unit cell of $Y Ba_2 Cu_3 O_7$ which is superconductor when T = 90K is a stack of alternating oxygen deficient (or defect) cubic pervoskite structure in the idealised sequence noted, arrows. The chains of alternating Ba : Y : Ba atoms which characterize the $Y Ba_2 Cu_3 O_7$ lattice are perpendicular o Cu-O planes with alternating oxygen defects (vacancies) as shown in below figure 1.22.

So the oxygen defects induce superconductivity in the pervoskite structures.

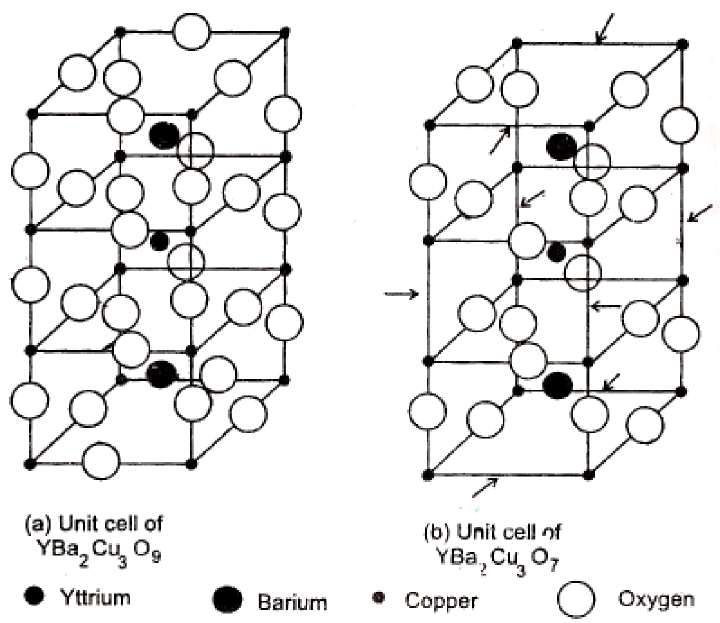


Fig 1.21 *Fig 1.22*

Fig 1.21 & Fig 1.22 Pervoskite ordinary and superconducting structures.